Supplement:
Beyond Success and Violation—Manipulation-Entry Collapse (Extended Materials)

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Abstract

This supplement accompanies the RA-L main text. It provides implementation details, additional replications, extended robustness figures, full phase-transition matrices, falsification checks, and data provenance tables.

Contents

1 Implementation Details

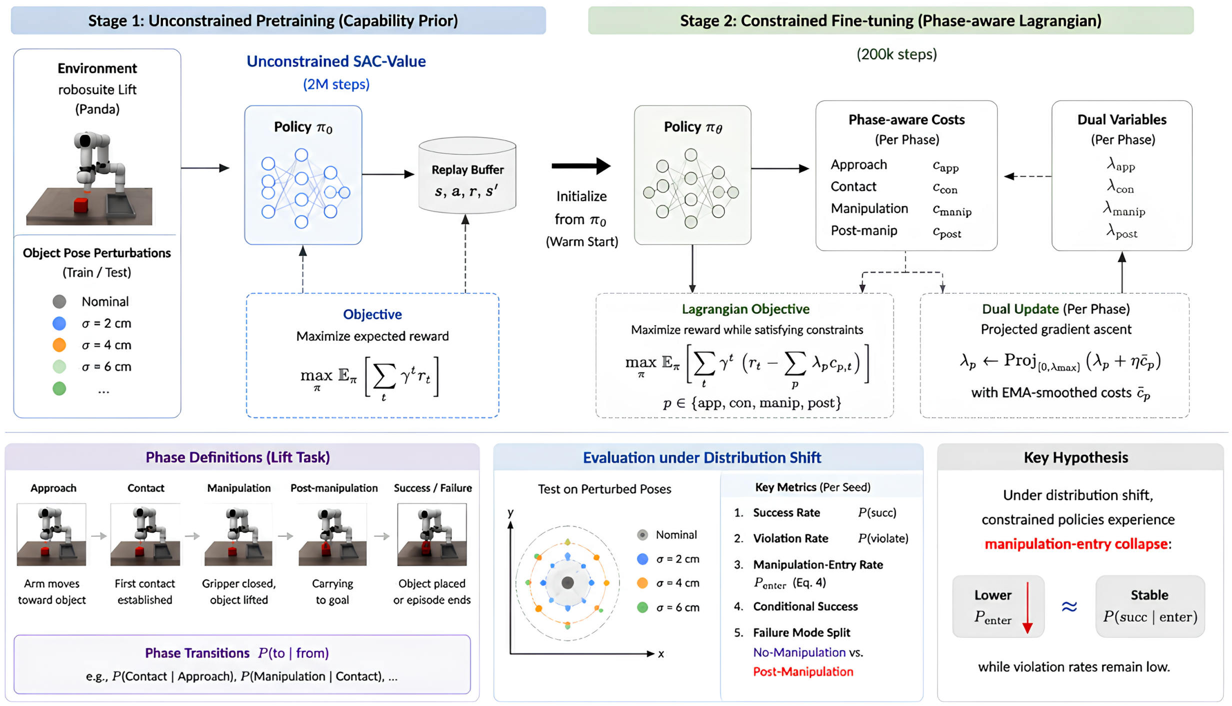

Figure 1: Detailed pipeline overview (companion to Fig. 2 in the main text). Stage 1: unconstrained SAC-Value pretraining (2M steps) on robosuite Lift. Stage 2: 200k-step phase-aware Lagrangian fine-tuning with per-phase costs c_p, dual variables λ_p, and EMA-smoothed force penalties. Bottom panels: phase timeline, evaluation metrics under pose perturbations (including manipulation-entry rate P_{enter}), and the entry-collapse hypothesis. Schematic phase labels (*approach/contact/manipulation/post*) map to the implementation phases *free/pre/contact/manip* used in the main text.

1.1 SAC-Value capability prior

The unconstrained prior uses twin Q-networks $Q_{\phi_i}(s, a)$, Gaussian policy $\pi_\theta(a|s)$, and state-value network $V_\psi(s)$. Critic targets follow the entropy-regularized Bellman backup,

$$y_Q = r + \gamma \left(\min_{i \in \{1,2\}} Q_{\phi_i'}(s', a') - \alpha \log \pi_\theta(a'|s') \right), \qquad a' \sim \pi_\theta(\cdot|s'), \tag{1}$$

with Q_{ϕ_i} trained by regressing to y_Q. The value network uses

$$y_V = r + \gamma V_{\psi'}(s'), \tag{2}$$

and the actor minimizes

$$\mathcal{L}_\pi^{\text{prior}} = \mathbb{E}_{s,a\sim\pi} \left[\alpha \log \pi_\theta(a|s) - \left(\min_i Q_{\phi_i}(s, a) - V_\psi(s) \right) \right]. \tag{3}$$

The prior is trained for 2M steps on Lift without safety constraints.

1.2 EMA force costs and phase limits

Let F_t denote gripper–object contact force. The smoothed force is

$$\bar{F}_t = \alpha_F \bar{F}_{t-1} + (1-\alpha_F) F_t, \qquad \alpha_F = 0.85. \tag{4}$$

Force costs activate on sustained exceedance of the safe threshold. Phase cost limits are $\epsilon_{\text{free}} = 0.02$, $\epsilon_{\text{pre}} = 0.03$, $\epsilon_{\text{contact}} = 0.08$, $\epsilon_{\text{manip}} = 0.12$.

1.3 Phase and global dual updates

Per-phase multipliers satisfy $\lambda_z = \exp(\log \lambda_z)$ and are updated from batch means $\bar{C}_z$ via

$$\mathcal{L}_\lambda = -\sum_z \log \lambda_z \left(\bar{C}_z - \epsilon_z\right), \qquad \lambda_z \leq \lambda_{\max}. \tag{5}$$

with $\lambda_{\max}$=20 in our runs. The global-λ arm uses the same cost pipeline with a single Lagrange multiplier instead of per-phase λ_z.

1.4 Interpretation Framework (Supplementary)

For readers who wish to connect the metrics to the training objective, we note the following interpretation framework (not a formal theorem). The M4.1 actor in the main text minimizes entropy-regularized advantage plus a phase-weighted cost penalty $\lambda_z C_z$, with sample expectations taken over on-policy rollouts. A complementary view is to track empirical phase occupancy $d^\pi(z)$ and manipulation-entry rate: in our experiments, reduced $d^\pi(\text{manip})$ under shift often co-occurs with stable $P(\text{succ} \mid \text{enter})$, which is consistent with changed interaction initiation rather than uniform post-contact degradation.

A matched global-λ arm shows a similar order of entry reduction to phase-aware M4.1 at σ=6 cm in our testbed (Table 6 in the main text). Fuller hypothesis–prediction mapping and alternative-explanation checks are in Tables 5 and 3 below.

1.5 Data provenance and reproducibility

Metrics in the main text are aggregated from evaluation and training logs in the project repository. Readers need not consult filenames to interpret the paper; this subsection maps cohorts and runs to paths for replication only.

Table 1: Evaluation log sources (repository-relative paths).

Cohort	Path
Main robustness grid	`robustness_m4_1/robustness_episodes.csv`
Three-policy attribution (σ=6 cm)	`robustness_attribution/robustness_episodes.csv`
Global-λ comparison	`robustness_sac_lag_warm_m4_1/robustness_episodes.csv`
Nominal M4.1 baseline / repl0	`multiseed_m4_1/eval_old_baseline_noseed.csv`, `multiseed_m4_1/eval_repl0.csv`
Nominal replications repl1–2	`multiseed_m4_1/eval_repl1.csv`, `multiseed_m4_1/eval_repl2.csv`
M4 warm-start nominal	`eval_cpc_m4_warm_sacv_200k.csv`

Table 2: Fine-tuning log sources for dual multipliers (Table 7, Figure 5). Each directory stores TensorBoard scalars and a JSON snapshot of final dual variables at 200k steps.

Run	Training log directory
Baseline	`cpc_m4_1_warm_sacv_logs/`
repl0	`cpc_m4_1_warm_sacv_repl0_logs/`
repl1	`cpc_m4_1_warm_prior_repl1_logs/`
repl2	`cpc_m4_1_warm_prior_repl2_logs/`

Table and figure regeneration scripts live in the project repository alongside these logs.

2 Supporting Evaluation Details

2.1 Nominal task success

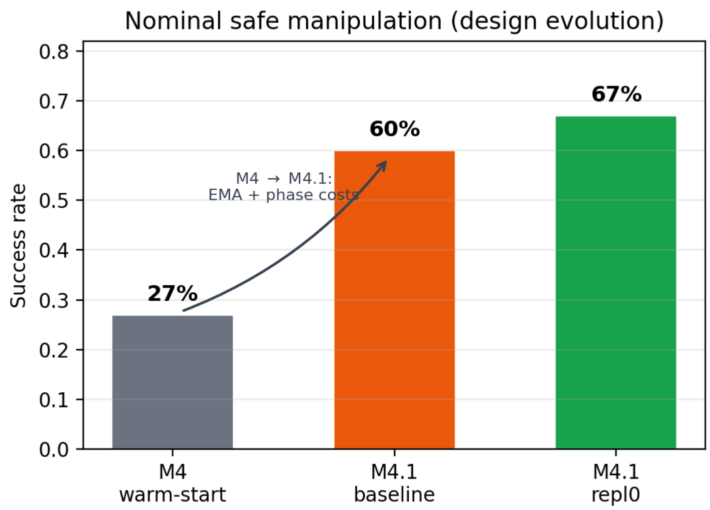

Figure 2: Nominal task success under the evaluation protocol (deterministic actions, unseeded resets; same cohort as Table 3). EMA phase costs (M4.1) are associated with higher success than instantaneous-force M4 warm-start (26.7% legacy eval vs. 60% baseline; repl0 67%).

2.2 Safety–robustness trade-off

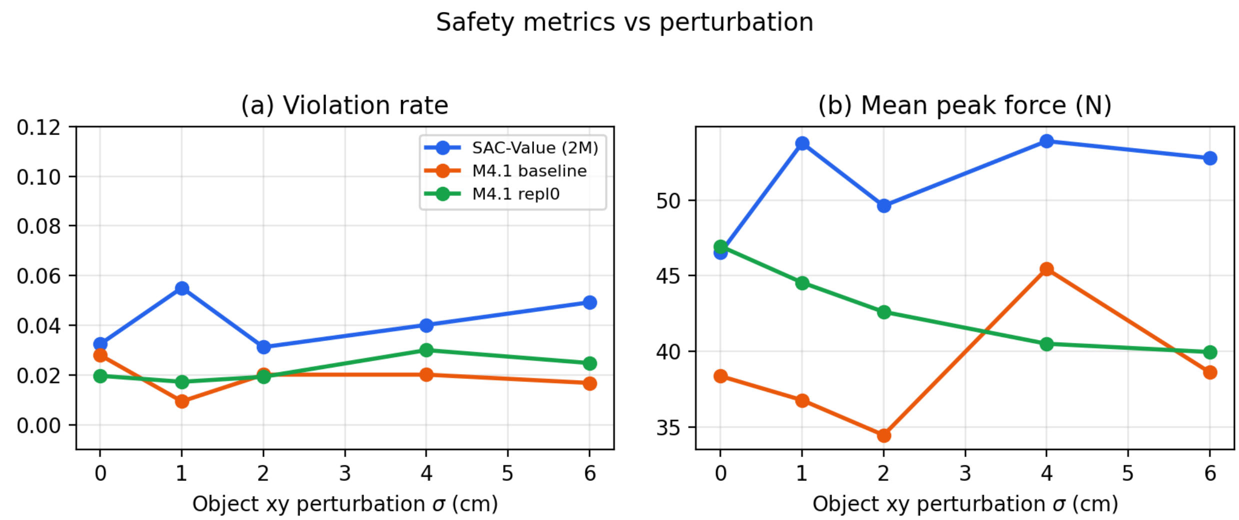

Figure 3: Violation rate and mean peak force vs. perturbation magnitude. At σ=6 cm, M4.1 keeps low violations (1.7–2.5%) while success falls, a pattern consistent with reduced manipulation entry rather than elevated post-contact violation in these runs.

At σ=6 cm, M4.1 baseline has 1.7% mean violation and 38.6 N mean peak force; repl0 has 2.5% and 40.0 N. The SAC-Value prior reaches higher success at the same perturbation but higher violation (4.9%) and force (52.8 N). Low violation therefore coexists with low success for constrained policies.

2.3 Training-time dual vs. manipulation occupancy

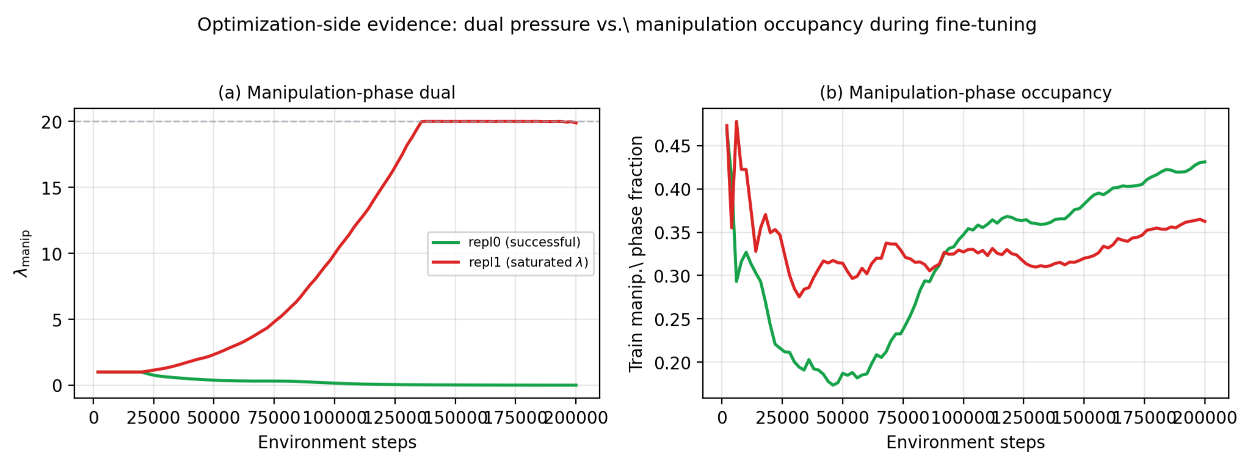

Figure 4: Training-time λ_{manip} vs. train-time manipulation-phase fraction for two M4.1 replications. repl0 reaches strong nominal success; repl1 saturates λ_{manip} near $\lambda_{\max}$=20 while manipulation fraction stays low.

Figure 4 complements Appendix Figure 5: successful fine-tuning can lower λ_{manip} while maintaining high manipulation occupancy; failed replications can sustain saturated dual pressure alongside reduced manipulation contact.

2.4 Approach behavior at σ=6 cm

Episode-level logs at σ=6 cm support a contact-avoidance reading of no-manipulation failures. Relative to the SAC-Value prior, M4.1 baseline episodes show larger mean end-effector–object distance (4.7 cm vs. 1.3 cm), higher pre-contact phase fraction (16% vs. 5%), and lower manipulation-phase fraction (38% vs. 67%) over 30 matched-reset episodes. These differences are consistent with more conservative approach and delayed commitment to manipulation under shift (prediction **P3**, Table 5).

2.5 M4 instantaneous-force ablation

Table 5 includes an M4 warm-start arm (instantaneous force thresholds, no EMA). At σ=6 cm, M4 reaches 23.3% success and 40.0% manipulation entry with 58.3% $P(\mathrm{succ} \mid \mathrm{enter})$, so both constrained variants show entry well below the prior (83.3%). At σ=0 cm, M4 and M4.1 share similar entry rates (73.3%) but M4 conditional success given entry collapses to 31.8% vs. 81.8% for M4.1, and M4 nominal success is only 23.3% vs. 60.0% for M4.1. EMA smoothing is therefore associated with stabilizing nominal constrained learning rather than with uniquely causing entry suppression under shift.

2.6 Legacy global-λ without warm-start

The global-λ fine-tuning run without SAC-Value warm-start fails to learn manipulation-capable behavior (near-zero manipulation entry at σ=0 and σ=6 cm). It serves as a negative control: a capable prior is required before global-λ fine-tuning can be compared fairly to phase-aware M4.1.

3 Alternative Explanations and Falsification Checks

We examined whether manipulation-entry collapse could instead be explained by EMA smoothing, warm-start skill loss, $\lambda_{\max}$ saturation, or phase-cost architecture alone. Table 3 summarizes the checks; overall evidence remained consistent with the manipulation-entry interpretation rather than with any single alternative alone.

Table 3: Alternative explanations considered and evidence against them (or scope limits) in the Lift testbed.

Alternative	Why unlikely / bounded	Evidence
EMA smoothing alone causes collapse	M4 (no EMA) shows lower entry than SAC-Value at σ=6 cm	Table 5
Warm-start removes manipulation skill	High $P(\mathrm{succ} \mid \mathrm{enter})$ for M4.1	82.4% vs. 84.0% at σ=6 cm
$\lambda_{\max}$ too small	repl1 saturates without recovering entry	Appendix Table 7, Fig. 5
Phase-cost *architecture* alone	Global-λ matches entry scale at σ=6 cm	Table 6, p=1.0

Table 4: Fisher exact tests on manipulation-entry rate at σ=6 cm (30 episodes per policy; two-sided).

Comparison	Entry rates	p-value
SAC-Value vs. M4.1	86.7% vs. 53.3%	$p = 0.010$
SAC-Value vs. global λ	86.7% vs. 53.3%	$p = 0.010$
M4.1 vs. global λ	53.3% vs. 53.3%	$p = 1.000$

4 Hypothesis–Prediction Map

Table 5 provides an optional mapping between interpretation hypotheses H1–H3 (not theorems) and empirical checks P1–P4 in this testbed.

Table 5: Hypothesis–prediction map (Lift testbed; H1–H3 not theorems).

Hyp.	Pred.	Signature	Status
H1	**P1**	Lower d^{π}(manip), reduced phase flow	Figs. 5, 4; supported
H1	**P2**	Stable $P(\text{succ} \mid \text{enter})$	Table 5; supported
H1	**P3**	Conservative approach before entry	Appendix 2.4; consistent
H2	**P4**	Global-λ same-order entry suppression	Tables 6, 4; supported
H3	—	Scope boundary	Section 6.4

5 Replication Variance

Table 6: Additional M4.1 replications under the legacy training and evaluation protocol. These runs document optimization variance and are not used to claim consistent 60–70% success across all replications.

Run	Success	Reward	Manip. frac	Violation	$F_{\max}$
repl0	66.7%	1618.2	0.731	2.45%	53.5 N
repl1	33.3%	430.2	0.359	2.27%	58.4 N
repl2	36.7%	952.8	0.394	1.37%	39.6 N
Mean ± std	$45.6 \pm 18.4\%$	1000.4 ± 595.4	0.495 ± 0.205	$2.03 \pm 0.58\%$	50.5 ± 9.7 N

Replication duals are not monotone with eval success alone. repl1 (lower success) saturates λ_{manip}; repl2 shows elevated λ_{free} with low λ_{manip}; repl0 succeeds nominally while λ_{free} sits at the cap and λ_{manip} stays small. These results support the interpretation that optimization instability is a central challenge in phase-aware constrained manipulation, rather than evidence that the nominal capability result is a single impossible event.

Table 7: Final per-phase Lagrange multipliers at the end of 200k-step fine-tuning (training-termination logs; Appendix 2). repl1 saturates λ_{manip}; repl2 shows elevated λ_{free} with low λ_{manip}; repl0 reaches strong nominal success (Table 6) despite λ_{free} at the cap and a small λ_{manip}.

Run	λ_{free}	λ_{pre}	λ_{contact}	λ_{manip}
repl0	20.0	1.5×10^{-8}	2.2×10^{-8}	0.0014
repl1	0.021	1.5×10^{-8}	3.0×10^{-8}	19.88
repl2	6.13	1.5×10^{-8}	2.1×10^{-8}	0.10

6 Training-Time Lagrange Multipliers

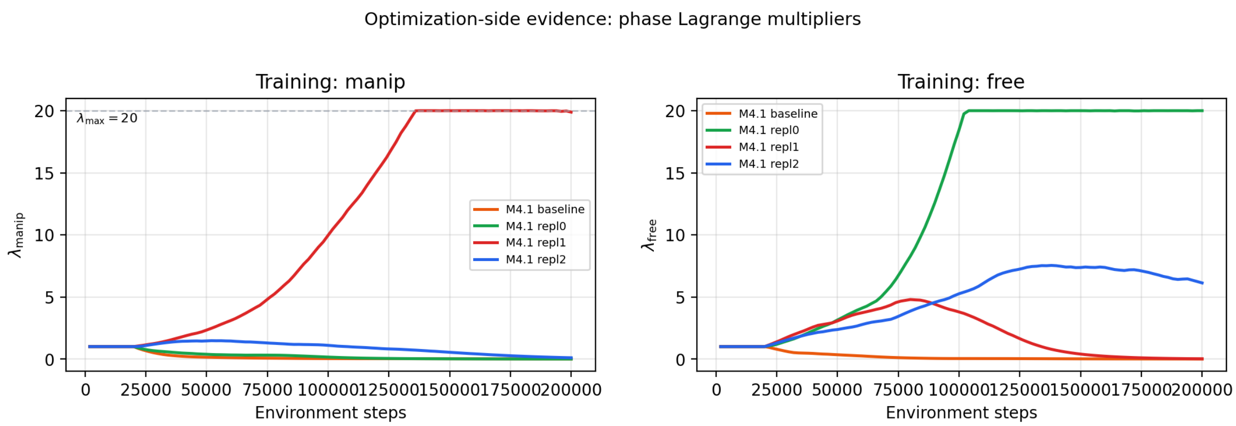

Figure 5: Training-time phase Lagrange multipliers for baseline, repl0, repl1, and repl2. **Left:** λ_{manip} (repl1 saturates near $\lambda_{\max} = 20$). **Right:** λ_{free} (repl2 remains elevated for repl2; repl0 also reaches the $\lambda_{\max}$ ceiling in training despite strong nominal evaluation).

This appendix complements the evaluation-side mechanism analysis with optimization-side evidence. We log phase-specific Lagrange multipliers λ_{free} and λ_{manip} during M4.1 fine-tuning from TensorBoard scalars for the baseline and three replication runs (Appendix 2). The manipulation multiplier is capped at $\lambda_{\max} = 20$ in our implementation.

Figure 5 shows qualitatively different dual trajectories across seeds. repl1 drives λ_{manip} to the upper bound while task success remains low (33.3% in Table 6). repl2 shows elevated λ_{free}, matching its higher free-space occupancy and reduced manipulation fraction in evaluation. repl0 does not saturate λ_{manip} but still reaches the λ_{free} cap at training end (Table 7), so dual pressure and nominal success are not one-to-one in our runs. These patterns support treating dual instability as part of the safe manipulation problem rather than as a mere implementation detail.

7 Phase-Aware vs. Global-λ Entry Disagreement at σ=6 cm

Table 6 in the main text reports identical manipulation-entry rates (53.3%, 16/30; Wilson CI $[36.1, 69.8]$) for phase-aware M4.1 and global-λ warm-start because Wilson intervals depend only on the entry count (k, n), not on which episodes enter. The two policies are evaluated on the same 30 matched reset seeds; entry outcomes still differ on 12 seeds (Table 8), so the aggregate tie is not a duplicate rollout artifact.

Table 8: Matched-reset episodes at σ=6 cm where phase-aware M4.1 and global-λ warm-start disagree on manipulation entry (threshold: manipulation fraction >0.05). Both policies still enter on 16/30 episodes overall (18/30 seeds agree); this table lists the remaining 12 seeds (6 phase-only, 6 global-only). Source logs: Appendix 1.

Seed	Phase λ_z entry	Global λ entry	Phase success	Global success
1	Yes	No	No	No
2	Yes	No	Yes	No
6	No	Yes	No	Yes
10	No	Yes	No	No
12	No	Yes	No	No
16	No	Yes	No	Yes
17	Yes	No	No	No
19	Yes	No	Yes	No
23	No	Yes	No	Yes
25	No	Yes	No	No
26	Yes	No	Yes	No
29	Yes	No	No	No

8 Additional Robustness and Phase-Transition Figures

The following figures supplement the main-text robustness and phase-transition analyses. Figure 6 overlays manipulation-entry rates for the SAC-Value prior, phase-aware M4.1, and global-λ warm-start constrained policies (Wilson 95% CI bands). Figure 7 provides the full metric grid; Figure 8 shows complete 4×4 transition matrices at $\sigma = 6$ cm.

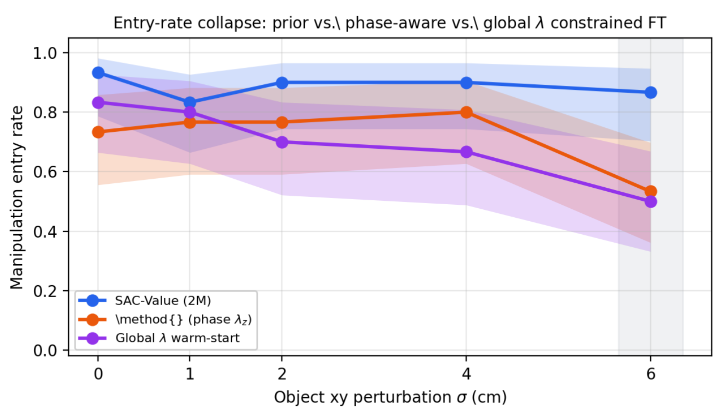

Figure 6: Manipulation-entry rate vs. object xy perturbation for three constrained-FT variants under a shared SAC-Value prior and M4.1 costs (30 episodes per (policy, σ); Wilson 95% CI). Phase-aware M4.1 and global-λ warm-start show a similar order of entry reduction at $\sigma = 6$ cm in this testbed.

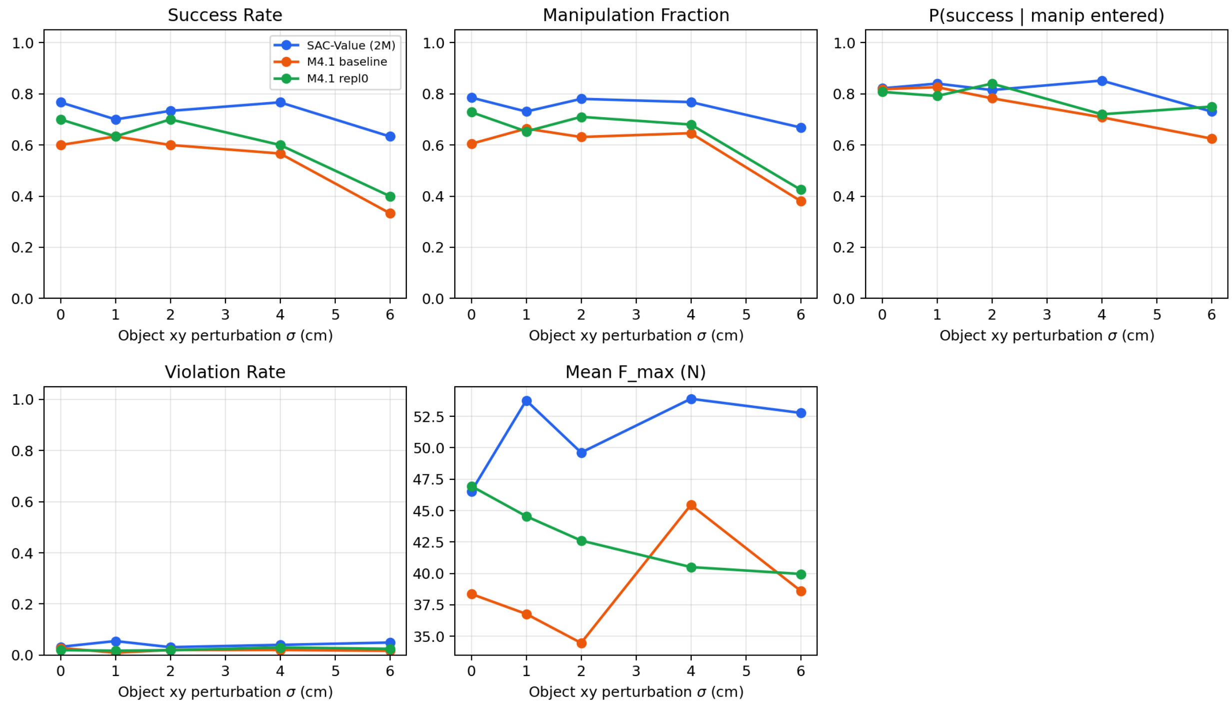

Figure 7: Full robustness grid with supplementary metrics.

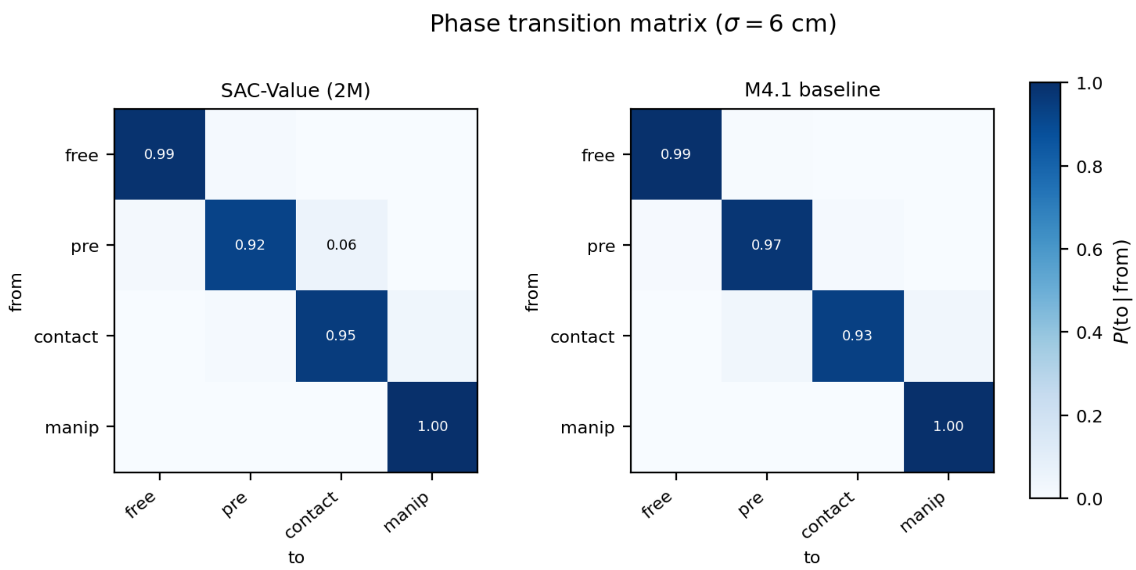

Figure 8: Full phase-transition matrices $P(\text{to}\,|\,\text{from})$ at $\sigma = 6$ cm (SAC-Value vs. M4.1 baseline). Supplement to Figure 5 in the main text.

References